

Finding the area of one face of a cube and of a cuboid

Six faces in three equal pairs — and what a single one of them is worth.

LESSON 143–145

3 × 40 MIN

MATEMATIKA 6, §30

S7 15 DISTANCE, AREA AND VOLUME

LESSON CLOCK

25'

30'

35'

25'

5

The three pairs of faces	25 min
One face of a cube	30 min
A face from the volume	35 min
Which faces the question needs	25 min
Homework	5 min

LEARNING OBJECTIVES

- Name the three different faces of a cuboid and find each area.
- Find one face of a cube from its total surface area.
- Find a face area from the volume and the perpendicular edge.
- Choose which faces a practical question actually needs.

TEXTBOOK

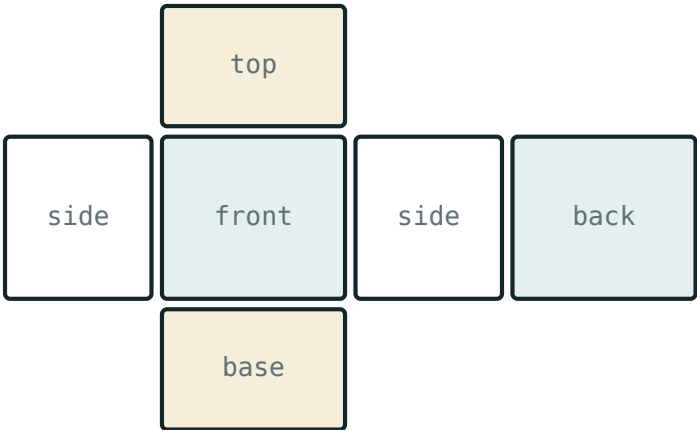
National	pp. 414–423
Cambridge	Stage 7, pp. 160–166
Workbook	Exercise 15.3

01

Explanation

The three pairs of faces

A cuboid with edges a , b , c has only three different faces — ab , bc and ac — and each of them appears twice, on opposite sides.



surface area = the area of the whole net

The net of a cuboid: three pairs of equal rectangles

CUBOID	AB	BC	AC	TOTAL
$5 \times 4 \times 3$	20 cm^2	12 cm^2	15 cm^2	94 cm^2
$8 \times 5 \times 2$	40 cm^2	10 cm^2	16 cm^2	132 cm^2
$10 \times 6 \times 4$	60 cm^2	24 cm^2	40 cm^2	248 cm^2
$6 \times 4 \times 3$	24 cm^2	12 cm^2	18 cm^2	108 cm^2

THE LARGEST FACE USES THE TWO LARGEST EDGES

In the $5 \times 4 \times 3$ cuboid the largest face is $5 \cdot 4 = 20$ and the smallest is $4 \cdot 3 = 12$. Pairing the edges off in the three possible ways is the whole exercise.

One face of a cube

All six faces of a cube are the same square, so the total is six of them and one of them is the total divided by six.

$one\ face = \frac{S}{6}$ and a = the square root of that

TOTAL SURFACE	ONE FACE	EDGE
54 cm^2	9 cm^2	3 cm
96 cm^2	16 cm^2	4 cm
150 cm^2	25 cm^2	5 cm
216 cm^2	36 cm^2	6 cm
294 cm^2	49 cm^2	7 cm

DIVIDE BY SIX BEFORE TAKING ANY ROOT

The square root of 96 is not the edge of anything here. $96 \div 6 = 16$ is a face, and 4 is the edge. Two steps, always in that order.

A face from the volume

Volume is a face multiplied by the edge standing perpendicular to it. Reverse that and a face area falls out of a division.

$$face = \frac{V}{\text{perpendicular edge}}$$

VOLUME	EDGE GIVEN	FACE PERPENDICULAR TO IT
120 cm ³	height 6 cm	20 cm ²
240 cm ³	height 4 cm	60 cm ²
84 cm ³	height 4 cm	21 cm ²
360 cm ³	height 5 cm	72 cm ²

THE UNITS CONFIRM THE CHOICE

cm³ ÷ cm = cm² — a face. Dividing by an area would have given a length instead, which is the previous topic, not this one.

Which faces the question needs

Practical questions rarely want all six faces. Read what is being covered before adding anything up.

TASK	FACES COUNTED	WORKING	ANSWER
wrapping a 20 × 15 × 10 cm box	all six	2(300 + 150 + 200)	1300 cm ²
card for an open box 8 × 5 × 4 cm	five — no lid	40 + 64 + 40	144 cm ²
painting the walls of a 5 × 4 × 3 m room	four walls	2(15) + 2(12)	54 m ²
the same room with the ceiling	four walls and the top	54 + 20	74 m ²

A FLOOR IS NOT PAINTED, A LID MAY NOT EXIST

Each of these questions has the same four numbers in it and four different answers. The reading, not the arithmetic, is what is being tested.

02 Worked examples

EXAMPLE 1 Find the three different face areas of a $5 \times 4 \times 3$ cm cuboid, and the total surface area.

① Pair the edges: $5 \cdot 4 = 20$, $4 \cdot 3 = 12$, $5 \cdot 3 = 15$.

② Each appears twice.

③ $2(20 + 12 + 15) = 94 \text{ cm}^2$.

The largest face uses the two largest edges ✓

ANSWER

20, 12 and 15 cm^2 ; total 94 cm^2

EXAMPLE 2 A cube has surface area 294 cm^2 . Find the area of one face and the edge.

① $294 \div 6 = 49 \text{ cm}^2$ for one face.

② That face is a square, so the edge is its square root.

③ 7 cm.

Check: $6 \cdot 49 = 294$ ✓

ANSWER

49 cm^2 and 7 cm

EXAMPLE 3 How much card is needed for an open box (no lid) 8 cm by 5 cm by 4 cm deep?

- ① Base: $8 \cdot 5 = 40 \text{ cm}^2$.
- ② Two long walls: $2 \cdot 8 \cdot 4 = 64$. Two short walls: $2 \cdot 5 \cdot 4 = 40$.
- ③ $40 + 64 + 40 = 144 \text{ cm}^2$.
The closed box would need 184 — one lid more.

ANSWER**144 cm²****03** **Interactive model**

Open out a cardboard box along its edges before the lesson; laying the net flat makes the three pairs visible in a way no drawing does.

Which face, and how big

The largest face of a $5 \times 4 \times 3$ cuboid is:

The smallest face of that cuboid is:

A cube of surface area 150 cm^2 has one face of:

That cube has edge:

25

5

6

12.5

A cuboid of volume 240 cm^3 and height 4 cm has base:

60 cm^2

960 cm^2

60 cm

24 cm^2

Wrapping a $20 \times 15 \times 10$ cm box needs:

650

1300

3000

900

An open box $8 \times 5 \times 4$ cm needs:

184

144

104

40

The four walls of a $5 \times 4 \times 3$ m room measure:

54 m²

74 m²

94 m²

60 m²

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	U'ZBEKCHA	РУССКИЙ
Face	Yoq	Грань
Opposite faces	Qarama-qarshi yoqlar	Противоположные грани
Base face	Asos yog'i	Основание
Side face	Yon yoq	Боковая грань
Lateral surface	Yon sirt	Боковая поверхность
Total surface area	To'la sirt yuzasi	Полная площадь поверхности
Net	Yoyilma	Развёртка
Open box	Qopqog'i yo'q quti	Открытая коробка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A cuboid has how many *different* face areas?

One face of a cube is the total surface:

The area of a face of an $8 \times 5 \times 2$ cuboid cannot be:

Volume divided by an edge gives:

a length

an area

a volume

nothing

An open box needs how many faces of card?

4

5

6

3

Painting only the walls of a room leaves out:

the walls

the floor and ceiling

one wall

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The largest face of a $5 \times 4 \times 3$ cm cuboid

ANSWER 20 cm^2

2. The smallest face of that cuboid

ANSWER 12 cm^2

3. The three faces of an $8 \times 5 \times 2$ cm cuboid

ANSWER $40, 16$ and 10 cm^2

4. One face of a cube of edge 4 cm

ANSWER 16 cm^2

5. One face of a cube of surface area 96 cm^2

ANSWER 16 cm^2

6. The number of faces of a cuboid

ANSWER 6

7. The faces of a cube are

ANSWER six equal squares

Medium · the standard question

7 PROBLEMS

1. One face of a cube of surface area 150 cm^2

ANSWER 25 cm^2

2. One face of a cube of surface area 294 cm^2

ANSWER 49 cm^2

3. The base of a cuboid of volume 120 cm^3 and height 6 cm

ANSWER 20 cm^2

4. The base of a cuboid of volume 240 cm^3 and height 4 cm

ANSWER 60 cm^2

5. The three faces of a $10 \times 6 \times 4 \text{ cm}$ cuboid

ANSWER $60, 40 \text{ and } 24 \text{ cm}^2$

6. The total surface of a $6 \times 4 \times 3 \text{ cm}$ cuboid

ANSWER 108 cm^2

7. The base of a cuboid of volume 360 cm^3 and height 5 cm

ANSWER 72 cm^2

Hard · stretch and reason

7 PROBLEMS

1. A cube of surface area 216 cm^2 : one face and the edge

ANSWER 36 cm^2 and 6 cm

2. The four walls of a 5 m by 4 m by 3 m room

ANSWER 54 m^2

3. The same room including the ceiling

ANSWER 74 m^2

4. Card for an open box $8 \times 5 \times 4 \text{ cm}$

ANSWER 144 cm^2

5. Paper to wrap a $20 \times 15 \times 10 \text{ cm}$ box

ANSWER 1300 cm^2

6. A cuboid has faces 12 , 15 and 20 cm^2 : its edges

ANSWER 3 , 4 and 5 cm

7. ...and its volume

ANSWER 60 cm^3

07 Homework

Homework — 5 tasks

For every task, say first which faces the question is asking about.

- List the three different face areas of a $9 \times 4 \times 2 \text{ cm}$ cuboid and find the total surface area.
- A cube has surface area 384 cm^2 . Find one face and the edge.
- A cuboid has volume 210 cm^3 and height 7 cm . Find the area of its base.
- How much card is needed for an open box 10 cm by 6 cm by 4 cm deep?
- A room is 6 m by 5 m by 3 m . Find the area of its four walls.